TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	75.0 ± 2.83
Correlation Stability	87.0 ± 1.63
Distribution Consistency	82.0 ± 6.38
Entropy Preservation	81.7 ± 0.47
Mean Z Score Drift	0.3 ± 0.09
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	39.3 ± 5.31
Data Density	98.4 ± 0.61
Coverage Score	56.3 ± 5.31
Feature Accuracy	84.3 ± 3.40
Novelty Score	0.6 ± 0.03

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Dataset Audit: parkinsons.csv

1. Dataset Portrait

Attribute	Value
Rows	195
Columns	23
Numeric features	23
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	3
Context % (*14)	40%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
MDVP:Fo	154.229	143.371	-10.857	7.0%	OK
MDVP:Fhi	197.105	168.985	-28.120	14.3%	Warn
MDVP:Flo	116.325	98.856	-17.469	15.0%	Warn
MDVP:Jitter	0.006	0.007	+0.000	4.5%	OK
MDVP:Jitter.1	0.006	0.007	+0.000	4.5%	OK
MDVP:RAP	0.003	0.003	+0.000	2.5%	OK
MDVP:PPQ	0.003	0.004	+0.000	7.5%	OK
Jitter:DDP	0.010	0.010	+0.000	1.6%	OK
MDVP:Shimmer	0.030	0.036	+0.006	20.4%	Warn
MDVP:Shimmer.1	0.030	0.036	+0.006	20.4%	Warn
Shimmer:APQ3	0.016	0.018	+0.003	17.3%	Warn
Shimmer:APQ5	0.018	0.022	+0.004	20.7%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	75.0 ± 2.83
Correlation Stability (*2)	87.0 ± 1.63
Distribution Consistency (*3)	82.0 ± 6.38
Entropy Preservation (*4)	81.7 ± 0.47
Mean Z Score Drift (*5)	0.3 ± 0.09
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	39.3 ± 5.31

Data Density (*8)	98.4 ± 0.61
Coverage Score (*9)	56.3 ± 5.31
Feature Accuracy (*10)	84.3 ± 3.40
Novelty Score (*11)	0.6 ± 0.03

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9743589743589743	0.7777777777777777	0.9658119658119659	-0.008547008547008517	
F1	0.980643539467069	0.8314339981006648	0.9738562091503269	-0.006787330316742095	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8717948717948718	0.7863247863247863	0.8461538461538461	-0.02564102564102566	
F1	0.9037242015450248	0.8316619463613759	0.8877896225852359	-0.01593457895978904	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9572649572649571	0.6324786324786325	0.9743589743589745	0.017094017094017106	
F1	0.9685551485551486	0.6733162461785506	0.9801307189542484	0.011575570399099807	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9487179487179486	0.8290598290598291	0.9316239316239315	-0.017094017094017106	
F1	0.9609674981103553	0.8753439153439153	0.9502729528535981	-0.010694545256757207	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7777777777777778	0.3931623931623931 5	0.5042735042735043	-0.2735042735042735	
F1	0.8614267676767677	0.3249327956989247 5	0.5763888888888888	-0.28503787878787884	

Model: LR

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8803418803418803	0.39316239316239315	0.7692307692307692	-0.11111111111111109	
F1	0.91232423490488	0.3650326797385621	0.8338831409006847	-0.07844109400419534	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8461538461538463	0.3675213675213675	0.7521367521367521	-0.09401709401709402	
F1	0.8918089248277927	0.2446963092124386	0.8136526366715046	-0.07815628815628817	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.717948717948718	0.4273504273504273	0.641025641025641	-0.07692307692307691	
F1	0.749906191369606	0.4587348217783000 5	0.6549707602339182	-0.09493543113568788	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7777777777777778	0.4017094017094018	0.7435897435897436	-0.03418803418803418	
F1	0.8627729426875721	0.3825344091124822	0.8189542483660132	-0.0438186943215589	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9401709401709403	0.7521367521367521	0.9145299145299145	-0.02564102564102563	
F1	0.9570703384745131	0.8282501124606387	0.9366396131102013	-0.020430725364311748	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	119.992	157.302	74.997	0.00784	0.00784	0.0037	0.00554	0.01109	0.04374	0.04374
2	122.4	148.65	113.819	0.00968	0.00968	0.00465	0.00696	0.01394	0.06134	0.06134
3	116.682	131.111	111.555	0.0105	0.0105	0.00544	0.00781	0.01633	0.05233	0.05233
4	116.676	137.871	111.366	0.00997	0.00997	0.00502	0.00698	0.01505	0.05492	0.05492
5	116.014	141.781	110.655	0.01284	0.01284	0.00655	0.00908	0.01966	0.06425	0.06425
6	120.552	131.162	113.787	0.00968	0.00968	0.00463	0.0075	0.01388	0.04701	0.04701
7	120.267	137.244	114.82	0.00333	0.00333	0.00155	0.00202	0.00466	0.01608	0.01608
8	107.332	113.84	104.315	0.0029	0.0029	0.00144	0.00182	0.00431	0.01567	0.01567
9	95.73	132.068	91.754	0.00551	0.00551	0.00293	0.00332	0.0088	0.02093	0.02093
10	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
11	88.333	112.24	84.072	0.00505	0.00505	0.00254	0.0033	0.00763	0.02143	0.02143
12	91.904	115.871	86.292	0.0054	0.0054	0.00281	0.00336	0.00844	0.02752	0.02752
13	136.926	159.866	131.276	0.00293	0.00293	0.00118	0.00153	0.00355	0.01259	0.01259
14	139.173	179.139	76.556	0.0039	0.0039	0.00165	0.00208	0.00496	0.01642	0.01642
15	152.845	163.305	75.836	0.00294	0.00294	0.00121	0.00149	0.00364	0.01828	0.01828
16	142.167	217.455	83.159	0.00369	0.00369	0.00157	0.00203	0.00471	0.01503	0.01503
17	144.188	349.259	82.764	0.00544	0.00544	0.00211	0.00292	0.00632	0.02047	0.02047
18	168.778	232.181	75.603	0.00718	0.00718	0.00284	0.00387	0.00853	0.03327	0.03327
19	153.046	175.829	68.623	0.00742	0.00742	0.00364	0.00432	0.01092	0.05517	0.05517
20	156.405	189.398	142.822	0.00768	0.00768	0.00372	0.00399	0.01116	0.03995	0.03995

B.1 Audit Trace: Fold 1 | Ctx: 40% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	113.4	133.344	107.816	0.00451	0.00451	0.00219	0.00283	0.00658	0.04879	0.04879
2	198.116	233.099	174.478	0.02714	0.02714	0.01568	0.01699	0.04705	0.08684	0.08684
3	96.106	108.664	84.51	0.00694	0.00694	0.00389	0.00415	0.01168	0.04024	0.04024
4	112.876	148.826	106.981	0.005	0.005	0.0027	0.00289	0.0081	0.03999	0.03999
5	173.898	211.35	74.677	0.00448	0.00448	0.00237	0.00254	0.0071	0.06727	0.06727
6	169.774	191.759	151.451	0.01568	0.01568	0.00863	0.00946	0.02589	0.08143	0.08143
7	120.5	135.2	110.3	0.005	0.005	0.0025	0.003	0.007	0.05	0.05
8	200.0	240.0	180.0	0.03	0.03	0.015	0.017	0.05	0.09	0.09
9	95.0	110.0	85.0	0.007	0.007	0.004	0.0045	0.012	0.04	0.04
10	115.0	150.0	108.0	0.005	0.005	0.0028	0.003	0.008	0.04	0.04
11	170.0	215.0	75.0	0.0045	0.0045	0.0025	0.0026	0.007	0.065	0.065
12	160.0	190.0	150.0	0.015	0.015	0.008	0.009	0.025	0.08	0.08
13	110.417	131.067	103.37	0.00784	0.00784	0.00352	0.00514	0.01056	0.03715	0.03715
14	208.083	253.792	91.802	0.00757	0.00757	0.00428	0.00428	0.01285	0.06725	0.06725
15	244.99	272.21	239.17	0.00451	0.00451	0.00279	0.00237	0.00837	0.01897	0.01897
16	228.969	239.541	113.201	0.00238	0.00238	0.00136	0.0014	0.00408	0.01745	0.01745

B.2 Audit Trace: Fold 2 | Ctx: 40% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	214.289	260.277	77.973	0.00567	0.00567	0.00295	0.00317	0.00885	0.01884	0.01884
2	118.747	123.723	109.836	0.00331	0.00331	0.00168	0.00171	0.00504	0.01043	0.01043
3	176.17	185.604	163.564	0.00369	0.00369	0.00205	0.00218	0.00616	0.01851	0.01851
4	138.145	197.238	81.114	0.00544	0.00544	0.00294	0.00327	0.00883	0.02791	0.02791
5	116.682	131.111	111.555	0.0105	0.0105	0.00544	0.00781	0.01633	0.05233	0.05233
6	243.439	250.912	232.435	0.0021	0.0021	0.00109	0.00137	0.00327	0.01419	0.01419
7	150.0	200.0	100.0	0.005	0.005	0.002	0.003	0.008	0.015	0.015
8	120.0	130.0	110.0	0.004	0.004	0.001	0.002	0.005	0.012	0.012
9	180.0	190.0	160.0	0.006	0.006	0.003	0.004	0.009	0.02	0.02
10	140.0	150.0	120.0	0.007	0.007	0.002	0.003	0.006	0.018	0.018
11	130.0	135.0	115.0	0.003	0.003	0.001	0.002	0.004	0.011	0.011
12	200.0	210.0	190.0	0.002	0.002	0.001	0.001	0.003	0.014	0.014
13	245.51	262.09	231.848	0.00235	0.00235	0.00127	0.00148	0.0038	0.01608	0.01608
14	115.38	123.109	108.634	0.00332	0.00332	0.0016	0.00199	0.0048	0.01503	0.01503
15	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
16	143.533	162.215	65.809	0.01101	0.01101	0.00647	0.00467	0.01941	0.05384	0.05384

B.3 Audit Trace: Fold 3 | Ctx: 40% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	155.358	227.383	80.055	0.0031	0.0031	0.00159	0.00176	0.00476	0.01718	0.01718
2	188.62	216.302	165.982	0.01719	0.01719	0.00996	0.01027	0.02987	0.07118	0.07118
3	170.756	450.247	79.032	0.00555	0.00555	0.00244	0.00261	0.00731	0.01725	0.01725
4	166.605	206.008	78.032	0.00742	0.00742	0.00387	0.00453	0.01161	0.0664	0.0664
5	136.969	166.607	66.004	0.00923	0.00923	0.00507	0.00463	0.0152	0.03111	0.03111
6	112.547	133.374	105.715	0.00355	0.00355	0.00166	0.0019	0.00499	0.01358	0.01358
7	145.678	210.345	85.123	0.0045	0.0045	0.0021	0.0023	0.0054	0.0182	0.0182
8	132.456	198.765	75.432	0.0067	0.0067	0.0035	0.0039	0.0091	0.0223	0.0223
9	167.89	230.456	90.123	0.0058	0.0058	0.0029	0.0031	0.0078	0.0194	0.0194
10	150.234	220.678	88.456	0.0039	0.0039	0.0018	0.002	0.0052	0.0176	0.0176
11	140.567	215.789	92.345	0.0041	0.0041	0.0024	0.0026	0.0065	0.0201	0.0201
12	155.678	240.123	95.678	0.005	0.005	0.0027	0.0029	0.0071	0.021	0.021
13	242.852	255.034	227.911	0.00225	0.00225	0.00117	0.00139	0.0035	0.01494	0.01494
14	139.224	586.567	66.157	0.03011	0.03011	0.01854	0.01628	0.05563	0.09419	0.09419
15	186.163	197.724	177.584	0.00298	0.00298	0.00165	0.00175	0.00496	0.01495	0.01495
16	116.879	131.897	108.153	0.00788	0.00788	0.00334	0.00493	0.01003	0.02645	0.02645